# Deep Learning Foundations

T1-02

DLS 2026

11 October 2026

# The learning chain

Examples $\rightarrow$ representation $\rightarrow$ prediction $\rightarrow$ loss $\rightarrow$ gradient $\rightarrow$ update. Every link can fail differently.

# A binary classifier

$z = w^T x + b$, $p = \sigma(z)$, $\ell = -[y \log p + (1 - y) \log(1 - p)]$. The loss translates a task judgment into an optimization signal.

# The useful derivative

$\partial \ell / \partial z = p - y$ and $\nabla_w \ell = (p - y)x$. Underprediction of a positive example raises its logit; overprediction of a negative lowers it.

# Backpropagation

Apply the chain rule backward through a computation graph. It efficiently reuses intermediate calculations; it does not guarantee a good task or data set.

## Worked update

For $x = (2, -1)$, $w = (.1, .2)$, $b = 0$, $y = 1$: $p = .5$, gradient $= (-1, .5)$; a small update increases this example's probability.

# Why training loss is insufficient

Leakage, nonrepresentative splits, imbalance, and a bad objective can all yield an attractive curve and a bad deployed system.

# Verification habits

Use held-out data; make preprocessing split-aware; inspect errors; finite-difference-check a small gradient implementation.

# Question

What would convince you that an observed accuracy gain is generalization rather than leakage or a shifted test distribution?